



北京邮电大学
Beijing University of Posts and Telecommunications

BUPT Compilers Lab3 2024Fall

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Container Number:

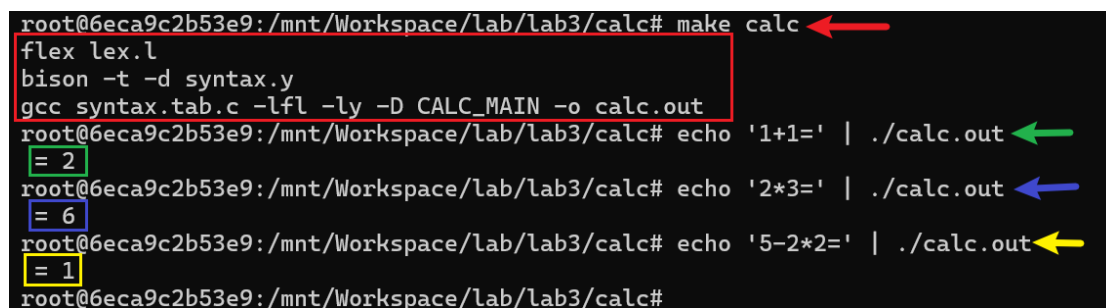
6eca9c2b53e9aaa0c65b6af6613814be05cc0d8ccc63b7df064c4a7230aab09f

1 Interacting Bison with Flex

The commands were used:

```
make calc  
  
echo '1+1=' | ./calc.out  
  
echo '2*3=' | ./calc.out  
  
echo '5-2*2=' | ./calc.out
```

The screenshot of the compilation and execution results.



A terminal window showing the compilation and execution of a calculator program. The commands and their outputs are as follows:

```
root@6eca9c2b53e9:/mnt/Workspace/lab/lab3/calc# make calc  
flex lex.l  
bison -t -d syntax.y  
gcc syntax.tab.c -lfl -ly -D CALC_MAIN -o calc.out  
root@6eca9c2b53e9:/mnt/Workspace/lab/lab3/calc# echo '1+1=' | ./calc.out  
= 2  
root@6eca9c2b53e9:/mnt/Workspace/lab/lab3/calc# echo '2*3=' | ./calc.out  
= 6  
root@6eca9c2b53e9:/mnt/Workspace/lab/lab3/calc# echo '5-2*2=' | ./calc.out  
= 1  
root@6eca9c2b53e9:/mnt/Workspace/lab/lab3/calc#
```

Colored arrows in the original image point to specific parts: a red arrow points to the 'make calc' command; a green arrow points to the output '= 2'; a blue arrow points to the output '= 6'; and a yellow arrow points to the output '= 1'.

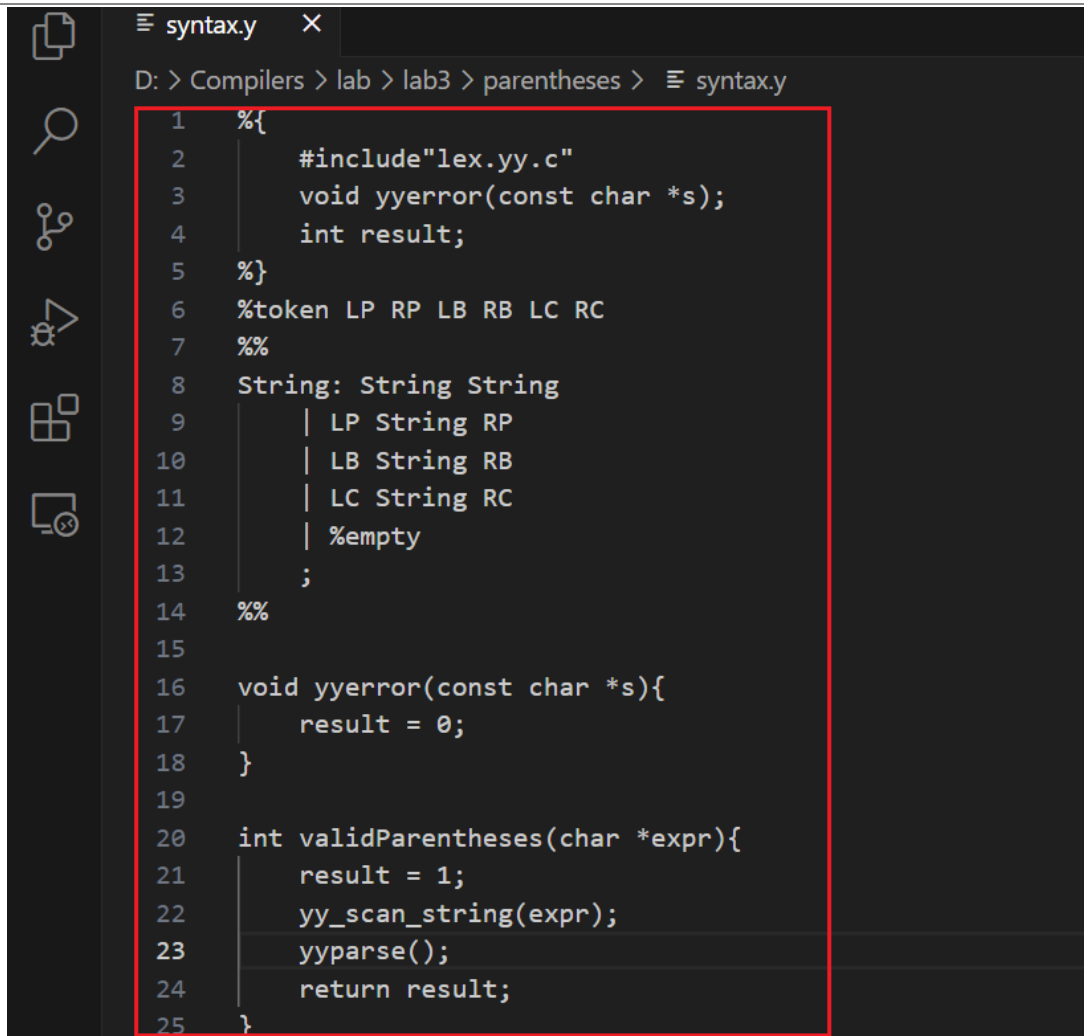
2 Validating parentheses

2.1 Code

The implementation of syntax.y is as follows.

```
%{  
    #include "lex.yy.c"  
    void yyerror(const char *s);  
    int result;  
}%  
  
%token LP RP LB RB LC RC  
%%  
  
String: String String  
    | LP String RP  
    | LB String RB  
    | LC String RC  
    | %empty  
    ;  
%%  
  
void yyerror(const char *s){  
    result = 0;  
}
```

```
int validParentheses(char *expr){  
    result = 1;  
    yy_scan_string(expr);  
    yyparse();  
    return result;  
}
```



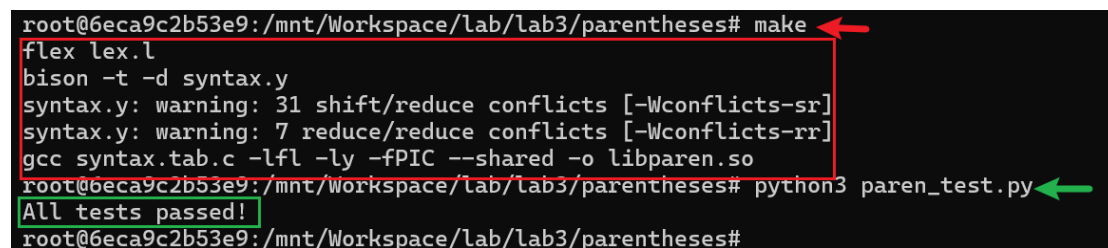
```
syntax.y  
D: > Compilers > lab > lab3 > parentheses > syntax.y  
1  %{  
2      #include "lex.yy.c"  
3      void yyerror(const char *s);  
4      int result;  
5  %}  
6  %token LP RP LB RB LC RC  
7  %%  
8  String: String String  
9      | LP String RP  
10     | LB String RB  
11     | LC String RC  
12     | %empty  
13     ;  
14  %%  
15  
16  void yyerror(const char *s){  
17      result = 0;  
18  }  
19  
20  int validParentheses(char *expr){  
21      result = 1;  
22      yy_scan_string(expr);  
23      yyparse();  
24      return result;  
25  }
```

2.2 Run

The commands were used:

```
make  
python3 paren_test.py
```

The screenshot of the compilation and execution results.



```
root@6eca9c2b53e9: /mnt/Workspace/lab/lab3/parentheses# make  
flex lex.l  
bison -t -d syntax.y  
syntax.y: warning: 31 shift/reduce conflicts [-Wconflicts-sr]  
syntax.y: warning: 7 reduce/reduce conflicts [-Wconflicts-rr]  
gcc syntax.tab.c -lfl -ly -fPIC --shared -o libparen.so  
root@6eca9c2b53e9: /mnt/Workspace/lab/lab3/parentheses# python3 paren_test.py  
All tests passed!  
root@6eca9c2b53e9: /mnt/Workspace/lab/lab3/parentheses#
```